

# Tokopedia: Building a powerful new address validation platform with Google Maps Platform

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Tokopedia improved delivery accuracy and checkout experience using Google Maps Platform to validate pick-up address and customer's delivery address.

## About Tokopedia

Tokopedia, one of Indonesia's leading tech companies with one of the country's largest marketplace platforms, serves more than 7.3 million merchants and more than 90 million active users every month.

Industries: Retail & Consumer Goods

Location: Indonesia

## Google Cloud results

- Improves delivery accuracy with precise location data on Google Maps
- Simplifies checkout process with autocomplete prompts in search bar
- Captures buyer and seller location data to improve exchange and company operations across logistics
- Provides more comprehensive data for third-party logistics partners

Precise location even standard address



## About OniGroup

OniGroup is an award-winning Google Cloud Partner specializing in location-based services, with offices in Asia, Australia, and the United Kingdom.



Google Maps Platform

(<https://cloud.google.com/maps-platform/>)

Geocoding API

(<https://developers.google.com/maps/documentation/geocoding/intro>)

Google Maps

(<https://cloud.google.com/maps-platform/maps/>)

Google Places API

(<https://cloud.google.com/maps-platform/places/>)

Indonesia's ecommerce industry is expected to hit \$65 billion by 2022. As online shopping becomes

more popular all over the country, it's set to accelerate in rural areas as well. One of the reasons?

Consumers outside of Java are saving 11% to 25% (<https://www.mckinsey.com/~/media/McKinsey/Featured%20Insights/Asia%20Pacific/The%20digital%20archipelago%20How%20online%20commerce%20is%20driving%20Indonesias%20economic%20development/The-digital-archipelago-Executive-summary.ashx>)

more by purchasing goods online instead of traditional retail.

A major player in this shift is the Indonesian technology company with the leading online marketplace, Tokopedia (<https://www.tokopedia.com/>).

With more than 90 million active users every month and more than 7 million merchants on its platform,

the company contributed to more than 1%

(<https://www.feb.ui.ac.id/en/blog/2019/10/19/lpem-feb-ui-tokopedia-berpengaruh-besar-pada-perekonomian-indonesia/>)

of Indonesia's economic growth in 2019.

As one of the fastest-growing technology companies in Asia, Tokopedia is driven by a mission to democratize commerce through technology.

Buyers can choose from more than 250 million products by shopping on Tokopedia's online marketplace. At the same time, the company helps merchants reach more markets - more than 97% of districts across Indonesia.

The customer journey on Tokopedia is only complete when the order is shipped from the merchant's store and delivered to the buyer's home. Thanks to a network of third-party logistics partners, Tokopedia delivers goods to anywhere in Indonesia.

In some cases, buyers can even request same-day delivery.

Because Indonesia is a geographically dispersed country covering more than 17,000 islands and spanning 3,200 miles, logistics outside of major urban areas can be a challenge. Imagine mountain roads that are accessible only on motorbikes or villages with no street numbers or names. On top of that, new suburban areas in large cities develop faster than the address system can keep up. In order to send the right logistics partner that can deliver on time and on cost, Tokopedia developed its own address validation platform, based on [Google Maps Platform](#) (/maps-platform) technology.

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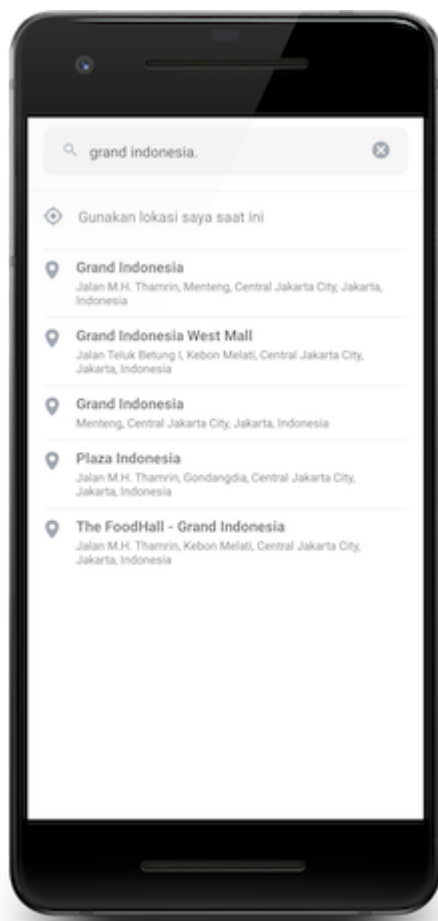
## **Helping users define their addresses to reduce delivery mix- ups**

Many rural households in Indonesia don't use standard postal addresses. It's common for homes and businesses in remote areas to use kampung (village) names or local landmarks instead of physical addresses. Unless a driver has local knowledge, they may struggle to find a location and waste precious time asking for directions.

"Whenever we had failed deliveries, the Tokopedia Care team members used to call customers to correct their address in our system," says Edwin Hartanto Mailoa, AVP of Logistics for Tokopedia. "Our new address validation platform, based on a combination of address Autocomplete and a visual mobile PIN on a Dynamic Map, allows the user to precisely define pickup and delivery addresses. By

validating the delivery address at the source, we can keep deliveries punctual and customers happy."

The Autocomplete feature of the [Places API](https://developers.google.com/places/web-service/intro) (<https://developers.google.com/places/web-service/intro>) suggests points of interest and location details such as street names and provinces as the customer starts to type in the address field. Each keystroke searches the Google Maps database in real time for accuracy, and recommends spelling corrections. Using address validation features, Tokopedia can prompt customers to enter the full address if any data is missing.



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When the user validates the address, they see a PIN appear on the map and they can move it to the precise location if needed. The [Geocoding API](https://developers.google.com/maps/documentation/geocoding/start) (https://developers.google.com/maps/documentation/geocoding/start)

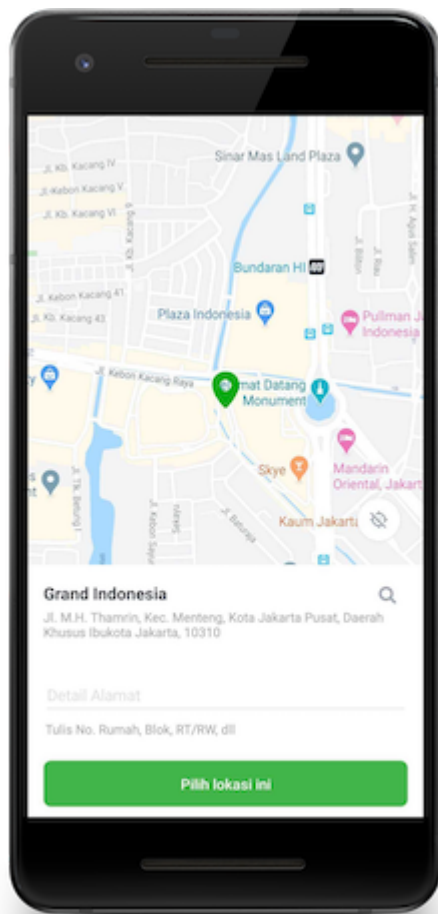
then converts the precise PIN location into geo-coordinates that Tokopedia shares with its logistics partners for routing purposes.

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Tokopedia uses the [Geolocation API](https://developers.google.com/maps/documentation/geolocation/intro) (https://developers.google.com/maps/documentation/geolocation/intro)

to determine the exact location of the user. If the customer allows location sharing with Tokopedia, the app picks up their current position by dropping an address pin along with the address on Google Maps. The customer then validates the displayed address with their location.





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## **Exact location data helps logistics partners find customers quickly**

Tokopedia worked with Google Cloud Partner OniGroup (<https://onigroup.com.au>) to optimize Google Maps Platform for logistics. “Successfully completed addresses have gone up by 10% since integrating address validation on Google Maps in our Tokopedia app, which is a great improvement,” says Mailoa.

"Moving forward, we're thinking to further enrich the platform by offering a satellite view, so a user could see the red roof of their house, and integrate this with potential pickup points. On top of that, exact location data for buyers and sellers will help us localize our offers, and run localized marketing campaigns and promotions more efficiently. With this, we'll be able to go beyond logistics," says Mailoa.

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can help your

business.